

Paper: 1MA1/2H					
Question		Answer	Mark	Mark scheme	Additional guidance
23	(a)	Shown	C1	for a method to find the area of half of the parallelogram or of the whole parallelogram, eg $\frac{1}{2}(2x - 1)(10 - x) \sin 150$ or $\frac{1}{2}(2x - 1)(10 - x) \times \frac{1}{2}$ oe or $(2x - 1)(10 - x) \sin 150$ or $(2x - 1)(10 - x) \times \frac{1}{2}$ oe	
			C1	for a correct expansion of the whole area eg $\frac{1}{2} (20x - 10 - 2x^2 + x)$ or $\frac{1}{2} (-2x^2 + 21x - 10)$ or $-x^2 + 10.5x - 5$	
			C1	complete chain of reasoning with fully correct algebra dealing with the inequality eg $x^2 - 10.5x + 5 < -15$ or $x^2 - 10.5x + 20 < 0$ or $2x^2 - 21x + 10 < -30$ which lead to $2x^2 - 21x + 40 < 0$	
	(b)	$2.5 < x < 8$	M1	for factorising, $(2x - 5)(x - 8)$	
			A1	for critical values, 2.5, 8	
			A1	for any statement that x is greater than 2.5 and x is less than 8	